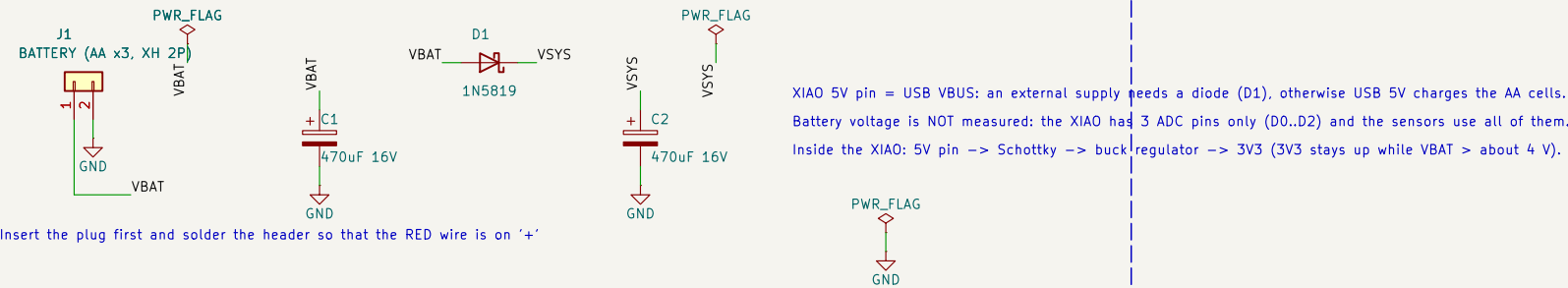


Linetracer2 Lite rev.L3 : XIAO ESP32C6 on pin headers, 3 x LBR-127HLD, 1 button, 1 buzzer, 6 RGB LEDs

Power: AA x3 (alkaline 4.5V recommended / NiMH 3.6V) -> VBAT (motors) ; VBAT -> D1 Schottky -> VSYS -> XIAO 5V pin. USB 5V can not back-feed the batteries. Never use 4 cells (the XIAO's 5V input clamps above about 6 V).

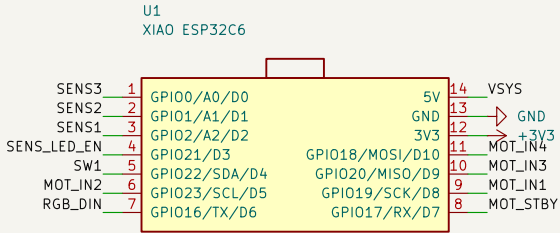
1. POWER



MECHANICAL: H1-H4 = gearbox frame M3 (x30/x70, y62/y96), H5 = front ball caster / skid, snap-in (x50, y11: behind the centre sensor)

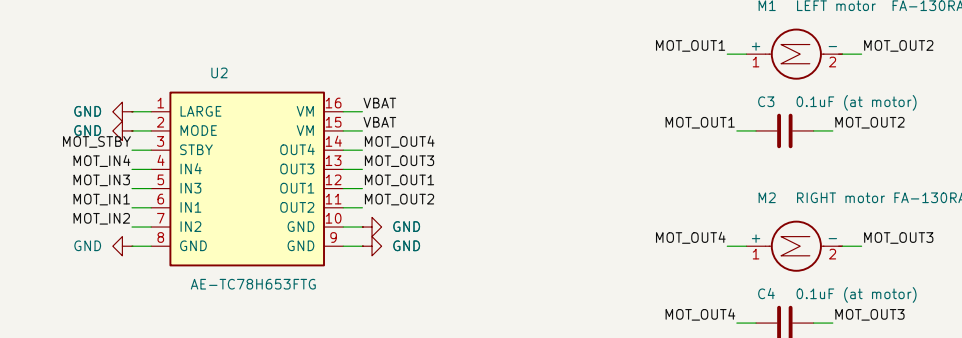


2. MCU Seeed Studio XIAO ESP32C6 (on two 1x7 pin headers)



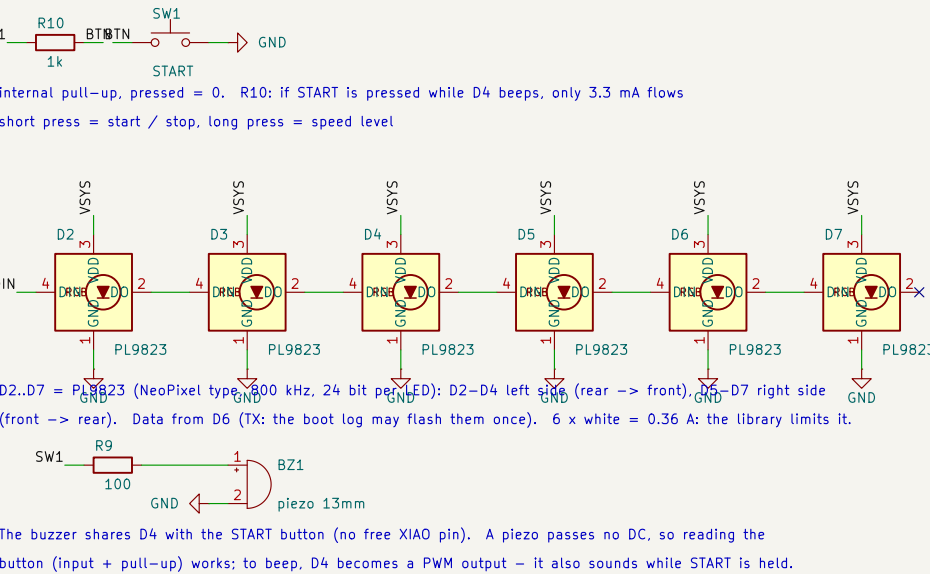
GPIO15 = yellow user LED on the XIAO (on = 0), a 2nd indicator for free. BOOT / RESET buttons are on the XIAO.
Reset state (ESP32-C6 datasheet): GPIO18-23 weak pull-up during reset, then floating;
GPIO16 (TX) / GPIO17 (RX) pull-up after reset + boot log / UART REPL -> no button and no motor input there.
On pin headers the XIAO sits 2.5 mm above the board: its bare bottom test pads can not touch the tracks.
Never solder a Li-ion cell to the XIAO's BAT pads on this robot (AA cells are on the 5V pin).

3. MOTOR DRIVER (Akizuki AE-TC78H653FTG module, DIP16)



LARGE=L : small mode (2 motors, 2A/ch) MODE=L : IN/IN control
IN1/IN2 -> OUT1/OUT2 = LEFT motor, IN3/IN4 -> OUT3/OUT4 = RIGHT motor (+ = OUT4)
H/L fwd, L/H rev, H/H brake, L/L coast. Over-current (ISD) latches OFF:
toggle STBY L->H to recover (done by the library).

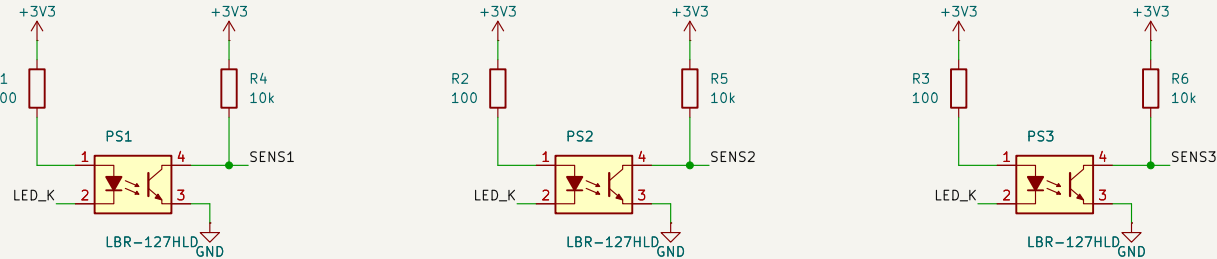
4. BUTTON / BUZZER / RGB LEDs



D2..D7 = PL9823 (NeoPixel type, 800 kHz, 24 bit per LED); D2-D4 left side (rear -> front), D5-D7 right side (front -> rear). Data from D6 (TX: the boot log may flash them once). 6 x white = 0.36 A: the library limits it.

5. LINE SENSORS 3 x LBR-127HLD (mount on the BOTTOM side, lens to the floor), 12 mm apart

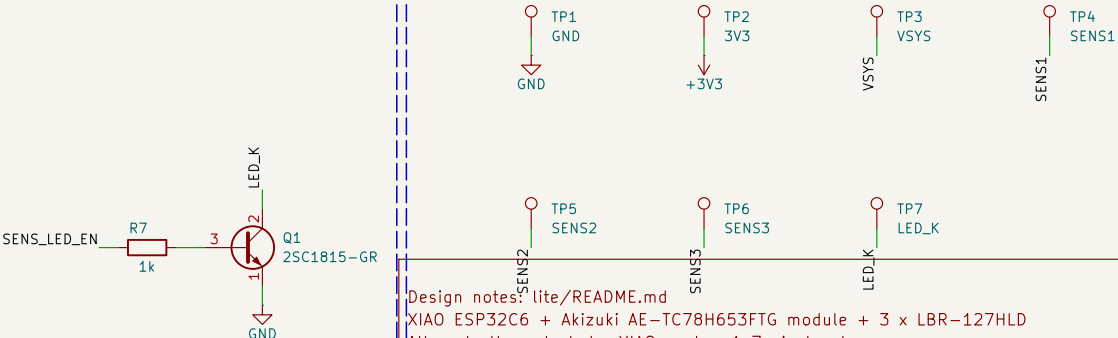
IR LED: (3.3V - 1.2V - 0.1V) / 100 ohm = about 20 mA each. Q1 switches all LEDs (ambient light cancel). Photo transistor: 10k pull-up -> white = low voltage, black = high voltage
The body is 5.6 mm tall: the front of the robot is lifted by the 7.5 mm ball caster or skid (lens about 2.25 mm above the floor)



PS1 = LEFT -> D2 (GPIO2/ADC), PS2 = CENTRE -> D1 (GPIO1/ADC), PS3 = RIGHT -> D0 (GPIO0/ADC) (seen from above, robot moving up the page)

Q1: all IR LEDs on when SENS_LED_EN = 1 (about 60 mA)

6. TEST PADS (multimeter, black probe on GND)



Design notes? lite/README.md		
XIAO ESP32C6 + Akizuki AE-TC78H653FTG module + 3 x LBR-127HLD		
All parts through-hole. XIAO on two 1x7 pin headers		
EDTC		
Sheet 1 of 1		
File: Linetracer2-Lite.kicad_sch		
3 LEDs (60 mA) about 0.1 V white (the IR LEDs are on (robot sensors leds(true)) about 2 V when off.		
Title: Linetracer2 Lite rev.L3 - XIAO ESP32C6 kids line tracer		
Size: A3	Date: 2026-10-07	Rev: L3
KiCad E.D.A. 10.0.6		Id: 1/1